

JIRAWAN HANPANOM : THE DEVELOPMENT OF FALL RISK PREDICTION MODELS FOR THE ELDERLY USING MACHINE LEARNING THESIS ADVISORS: ASST. PROF. PHICHAYASINI KITWATTHANATHAWON, Ph.D, AND AKKAPON WONGKOBLAB, Ph.D, 300 PP.

Elderly / Falls / Machine Learning / Prediction / Smart Insole Data

Falls are a significant and common issue among the elderly, posing a major public health concern and being a leading cause of injuries that often require hospitalization. Such incidents can lead to disability or even death. This research focuses on developing a machine learning-based model for predicting fall risk in older adults, following the CRISP-DM data mining process. The study utilizes gait data from elderly individuals wearing Surasole smart insoles and compares two data imbalance handling techniques, three feature selection methods, and twelve machine learning algorithms—resulting in a total of 744 modeling approaches. The objective is to identify the most effective and unbiased fall risk prediction model. The results show that the best-performing model is a Random Forest combined with backward feature selection using the Wrapper Method and Gradient Boosting Machines as well as Synthetic Minority Over-sampling Technique (SMOTE) for handling imbalanced data. This model achieved the highest accuracy, precision, and recall rates, all at 0.92. The final model will be integrated with gait analysis for real-time fall risk prediction in a mobile application, enabling healthcare professionals to provide timely and appropriate interventions.

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